



ARETA RETAIL ANALYTICS

From a blank page to a live analytics suite

How I built a fictional outdoor retailer, its data, a Power BI report and a web app, end to end

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Portfolio project · 2026

THE AIM

One retailer, three ways to see the same numbers

I wanted a portfolio piece that looks and works like a real retail analytics suite, built the way I'd build one at work: business questions first, then the data, the model and the reports.

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|-----------|-------------------------------|-----------|--------------------------|
| 01 | The idea and the brief | 05 | The web app |
| 02 | Inventing the business | 06 | Trusting the numbers |
| 03 | Building the data | 07 | Challenges and learnings |
| 04 | The Power BI model and report | 08 | Outcomes and what's next |



Why build it



01

Background

I work in retail merchandising and wanted to show the whole journey, not just a finished dashboard.



02

Objective

One synthetic retailer feeding a Power BI report, a web app and a SQL suite from the same data.



03

My role

Everything: the brand, the data generator, the model, the report, the web app and the docs.



04

Outcome

A live, documented project that answers real trading questions and shows how I work.

Meet Areta Mountain Systems

A fictional outdoor retailer with its own brand, stores and range. Real retail structure, none of it real.



The brand board I designed everything from

4

brands

361

stores and concessions

11

European markets

907

styles, 12,151 SKUs

3.9M

invoice lines

2023

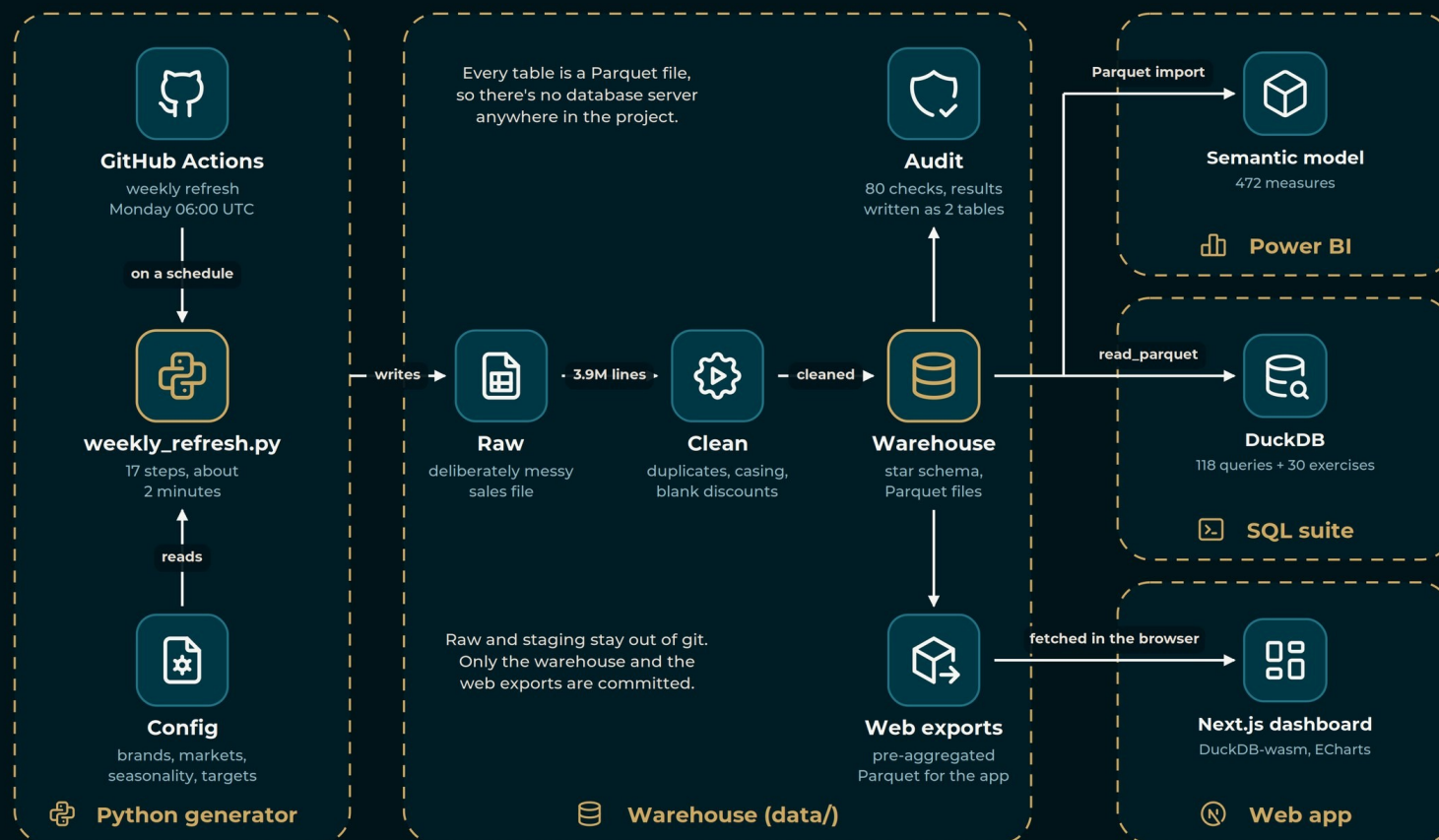
history starts in March

One pipeline in, three outputs out



How the data flows

One synthetic pipeline in, three ways of looking at the same numbers out



Everything upstream of the report is one Python pipeline, so the numbers can't drift between the three.



Python generator

17 steps in about two minutes, with a GitHub Action set up to run it every Monday.



Parquet star schema

No database server anywhere. Every table is a file in the warehouse.



Three outputs

Power BI, a Next.js web app and a DuckDB SQL suite all read the same files.

Making fake data behave like a real business



Foundations

A 364-day business calendar, stores, a four-brand product range, promotions, currencies and VAT.



Simulation

Sales at invoice-line level, with seasonality, discounts, multi-buys, returns and store traffic.



Realism rounds

Hero styles and a long tail, Black Friday and Christmas, growth that differs by area, and a web funnel.



Quality

A deliberately messy raw file, a cleaning step, and an 80-check audit at the end of every run.

1.00x → **1.5x**

Black Friday vs a normal Friday

Querying my own data in SQL showed Black Friday trading exactly like any other Friday. The flag was in the calendar, but nothing used it. Testing the data like a real business is what made it realistic.

A model built on patterns, not one-offs

472

DAX measures

26

tables in a star schema

2

calculation groups



Last year at any grain

One LY pattern moves the business year back and carries the period, week and weekday across, so it works by year, week or day.



Targets below period level

Targets are set per period. Spreading them across days by last year's weekday pattern makes weekly charts honest.



VAT and currency in one place

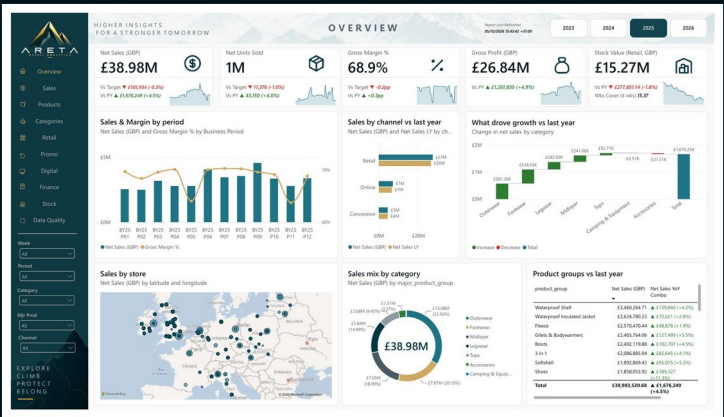
Calculation groups switch every money measure to include VAT or convert it, without a duplicate of each measure.



Stock as a balance

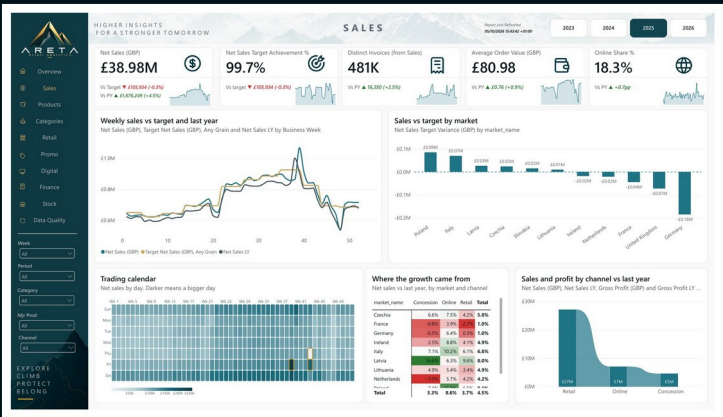
Stock takes the latest weekly snapshot instead of adding weeks together, and weeks of cover uses the recent rate of sale.

Ten pages, each answering one question



Overview

How is the business doing, and what drove it?



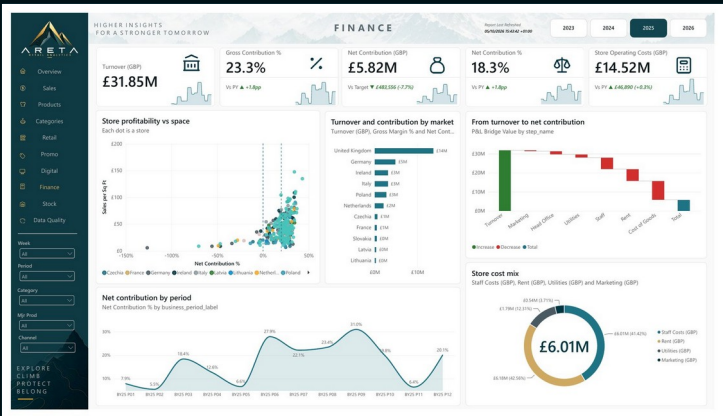
Sales

How are we trading, and where is the growth?



Digital

Where do shoppers drop out online?

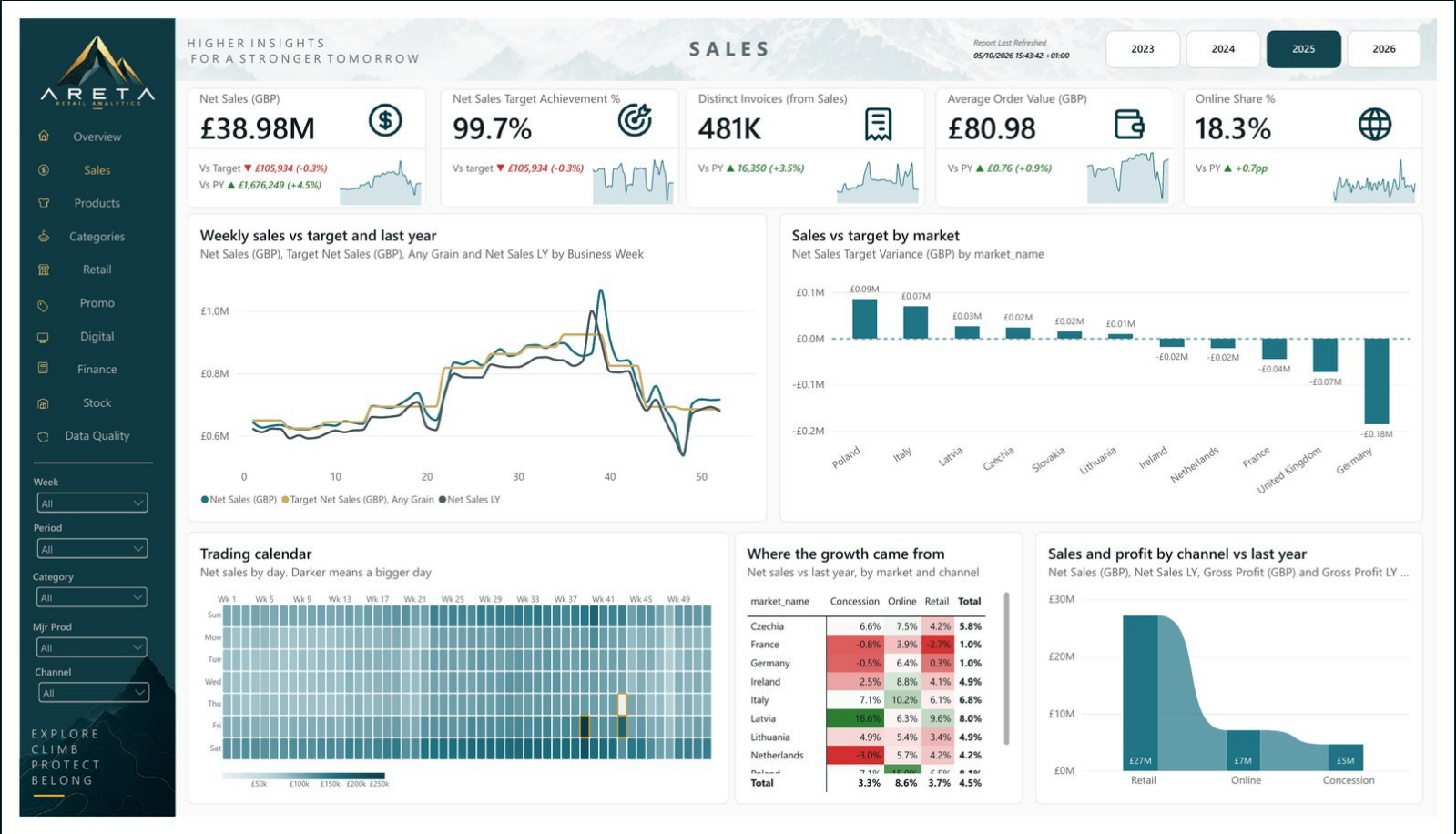


Finance

Where does the money go before profit?

Also: Products, Categories, Retail, Promo, Stock and Data Quality

Inside one page: Sales



KPI cards that explain themselves

Each card shows the value against last year and target, coloured by direction, with a weekly sparkline.



A trading calendar

A Deneb heatmap of every day in the year. Black Friday and Boxing Day stand out at a glance.



Where the growth came from

Growth by market and channel. Online grows everywhere; French stores go backwards.

The same numbers, in the browser



No server, no database

DuckDB runs inside the browser and reads the Parquet files directly.



Built for any screen

Phone drawer, tablet icon rail and a fit-to-screen desktop layout.



Filters in the URL

Every filtered view is a link you can bookmark or share.

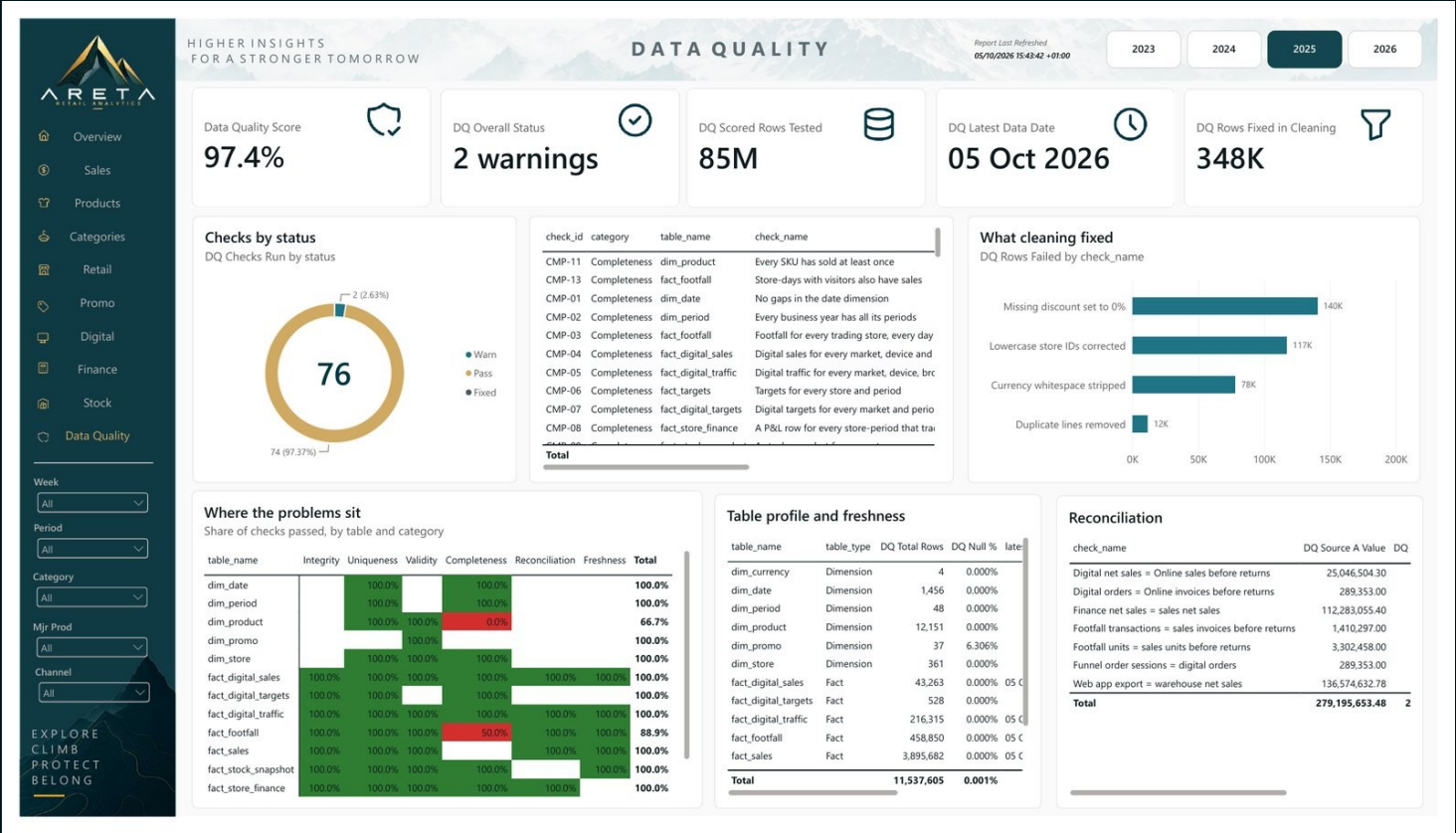


Live on Vercel

Eleven pages, built with Next.js and ECharts.

areta-retail-analytics.vercel.app

A report that checks its own data



97.4%

of scored checks pass: 74 of 76, with two known warnings I'd want flagged anyway

80 checks every run

7 reconciliations

348K rows fixed by cleaning

What went wrong, and what I learnt

CHALLENGE

Figures three to four times too high, but only with a slicer

Every year came out the same, so every variance read zero

Stock added four weeks together into one huge number

The web app worked in the build but failed in a real browser

WHAT I DID

Made the date-to-period relationship inactive and carried filters across with TREATAS

Added growth that differs by channel, category, market and store

Stock now takes the last snapshot in the filter

Fetches the file bytes and handed them to DuckDB directly

WHAT I LEARNT

Don't fight the relationship graph. Be explicit in the measure instead

Test data has to be able to move the measure you're testing

Know whether a measure is a flow or a balance before you sum it

Nothing replaces running it where your users will

Where it's landed, and where it goes next



10

Power BI pages



11

web app pages, live



118

SQL queries + 30 exercises



Weekly

refresh, scheduled in GitHub Actions

NEXT



Microsoft Fabric

A proof of concept: Lakehouse, Direct Lake and a scheduled pipeline



PL-300 and DP-600

Certifying what the project already shows



Report polish

VAT and currency toggles on the pages, and the funnel on the web app



ARETA

Thank you

Questions welcome

github.com/TGOSS1984/areta-retail-analytics
areta-retail-analytics.vercel.app